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Study of different Types of Network cables & crimping of cable with RJ45 connector

Aim: To study different types of network cables and understand the crimping process using an RJ45 connector

Concept: Network cables are used to connect computers and offer networking devices for data communication the most commonly used cable in LAN is the unshielded Twisted pair (UTP) cable. An RJ45 connector is attached to the ends of the cable of establish a network connection. The wires are arranged according to the 568A or 568B wiring standard before crimping & the completed cable is tested to ensure proper connectivity.

Types of Network Cables:-

	cable type	Description
1.	UTP [Unshielded Twisted Pair]	commonly Used for Ethernet LAN's & office nchoorks
2.	STP [shielded Twisted Pair]	Used in area with High electrical or electromagnetic interference
3.	Coaxial cable	Used for cable television broadbarrel, internet & CCTV systems
4	optical Fiber cable	Used For High-speed High Bandwidth & Long distance communication.

Crimping Procedure:

1. cut the UTP cable to the required length
2. strip about 2-3 cm of the outer insulation
3. Untwist & straighten the wires
4. Arrange the wires according to the T568B standard
5. Trim the wire ends evenly
6. Insert the wires into the RJ45 connector.
7. crimp the connector Using a crimping tool
8. Test the cable using a LAN cable Tester

Observation:

* The wires were Arranged according to the T568B Standard
* The RJ45 connector was crimped Successfully
* The cable tester indicated proper continuity
* The cable was ready for network communication.

Result: The different types of network cables were studied, and the UTP cable was Successfully terminated using and RJ45 connector and Tested for proper connectivity